

AREA III - COMPREHENSIVE EXAM

- lay, + by, + c
- 100 + 60
- Get*
2 = 100 - 100
0.5 = 100
- b* 1. According to Section 20 of RA 8749, which of the following practices is still allowed despite the general ban on incineration?
 - A. Regular operation of a municipal waste incinerator located in a public landfill
 - ✓* B. Backyard burning of dried leaves as part of traditional neighborhood sanitation
 - C. Routine incineration of biomedical waste from private hospitals
 - D. Burning of hazardous farm chemicals and synthetic packaging in open field
 - a* 2. A rice mill uses a wood-fired boiler and a corn dryer that emits smoke. Under the Clean Air Act (RA 8749), what must this facility secure to legally operate these stationary sources of air pollution?
 - ✓* A. A permit to operate from the DENR-EMB for its air pollution source.
 - B. A discharge permit from the DENR for air emissions.
 - C. An ordinance from the LGU allowing emissions
 - D. Nothing, since agricultural sources are exempt from air regulations.
 - b* 3. The Philippine Clean Water Act (RA 9275) regulates wastewater from agricultural and fisheries facilities. What permit is required for a facility that will discharge treated wastewater into a river?
 - ✓* A. Environmental Compliance Certificate from DENR
 - B. Discharge Permit from EMB
 - ✓* C. Water Rights Permit from NWRB
 - D. Import Permit for water treatment chemicals.
 - a* 4. Which agency is primarily tasked by RA 9275 (Clean Water Act) to oversee pollution from agricultural and aquaculture sources (non-point sources like farms and fishponds)?
 - A. DA
 - B. BFAR
 - ✓* C. DENR
 - D. LGU
 - b* 5. Prior to the enactment of the Clean Air Act and Clean Water Act, which law provided the broad guidelines for preventing pollution of air, water, and land in the Philippines?
 - ✓* A. RA 3931 - National Water and Air Pollution Control Commission Act
 - B. PD 984
 - C. RA 8749 - Clean Air Act of 1999
 - D. PD 1586 - EIS System Law
 - b* 6. RA 8749 provides certain exemptions to the general ban on waste incineration. Which of the following is one allowed practice despite the ban on incineration?
 - A. Burning biomedical and hazardous waste in an open pit. *×*
 - ✓* B. Traditional small-scale method of burning for yard waste
 - C. Incineration of municipal waste using any method, as long as it's done at night *×*
 - D. Uncontrolled burning of municipal garbage for volume reduction *×*
 - a* 7. The standard test for measuring the pollution potential of wastewater?
 - ✓* A. BOD₅
 - B. COD
 - C. pH
 - D. TS
 - c* 8. Which field of engineering involves timber extraction, reforestation practices, and the application of techniques that promote sustainable forest operations?
 - A. Agricultural and Biosystems Engineering
 - B. Environmental Engineering
 - ✓* C. Forest Engineering
 - D. Civil Engineering
 - c* 9. Which of the following is the correct sequence of steps in the timber extraction process?
 - A. Skidding, log processing, felling, transportation
 - B. Transportation, felling, skidding, log processing
 - ✓* C. Felling, log processing, skidding, transportation
 - D. Log processing, skidding, transportation, felling

10. What do you call this joint?

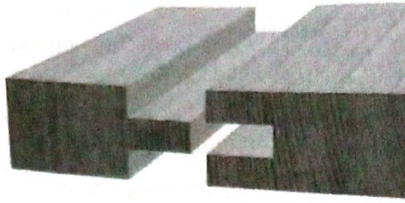


Image: bosch-professional.com

- A. Groove Joint
☒ B. Tongue and Groove Joint
 C. Box Joint
 D. Dado Joint
11. This is a term applied to wood after it is sawed or sliced into boards, planks, etc.
☒ A. Lumber
 B. Timber – piece of lumber five inches or 13 cm or larger in its smallest dimension
 C. Slab – rough lumber which is cut tangential to the annual rings
 D. Flitch – thick piece of lumber

12. A radial cracks originating at the heart of the logs commonly found in old trees.
☒ A. Heart shakes –
 B. Wind shakes – cracks or breaks across the annual rings of the wood during its growth caused by excessive bending
 C. Star shakes – composed of several heart shakes which radiate from the center of the log in star like manner
 D. Knots – occurs at the starting point of a limb or branch of the wood

13. Determine the total board foot of lumber which could be derived from a log 28 inches diameter by 20 feet long.
☒ A. 720 bd. ft.
 B. 702 bd. ft.
 C. 520 bd. ft.
 D. 502 bd. ft.

$$Bd. Ft. = \frac{(D-4)^2 \times L}{16} = \frac{(28-4)^2 \times 20}{16} = 720 \text{ bd. Ft.}$$

20. A rice mill operates 5 hours per day, 7 days per week, and processes 500 kg of rice per hour. If the emission factor for particulate matter (PM) is 5 lbs/ton and no pollution control facility is used, what is the estimated annual PM emissions?
- A. 0.91 tons/year
 B. 1.52 tons/year
☒ C. 2.06 tons/year
 D. 5.00 tons/year

Where D is in inches and L is in feet

14. What is the total board foot of 5 pieces of 2 in by 6 in by 14 ft wood?
 A. 50 bd. ft.
 B. 60 bd. ft.
☒ C. 70 bd. ft.
 D. 80 bd. ft.

$$bd. ft. = \frac{t \times w \times l}{12} = \frac{2 \times 6 \times 14}{12} = 14 \text{ bd. ft.}$$

Where t and w are in inches while l is in feet

15. CDM stands for:
☒ A. Clean Development Mechanism
 B. Climate Disaster Mitigation
 C. Carbon Disposal Management
 D. Conservation and Development Model

16. What convention creates CDM?
☒ A. Kyoto Protocol
 B. Paris Agreement
 C. Montreal Protocol
 D. Stockholm Protocol

17. What is a measure of the relative climate impact of a given greenhouse gas relative to the impact of carbon dioxide?

- A. Carbon Sequestration Index
 B. Emission Intensity Factor
☒ C. Global Warming Potential
 D. Atmospheric Residence Time

18. What is the global warming potential of methane?

- ☒ A. 21
 B. 72
 C. 1
 D. 10

19. According to the 2017 World Risk Report, what was the Philippines' global rank in terms of vulnerability to climate change?

- A. First
 B. Second
☒ C. Third
 D. Fourth

$$\frac{5 \text{ hrs}}{\text{day}} \times \frac{7 \text{ days}}{\text{week}} \times \frac{52 \text{ weeks}}{\text{yr}} = 182$$

$$500 \text{ kg} \times \frac{2.2 \text{ lbs}}{\text{kg}} \times \frac{5 \text{ hrs}}{\text{day}} \times \frac{7 \text{ days}}{\text{week}} \times \frac{52 \text{ weeks}}{\text{yr}} = 220 \text{ tons}$$

220 tons

5 lbs/ton

$$E = A \times EF \times \left(1 - \frac{ER}{100}\right)$$

Where E - emission estimation

A - activity rate

EF - emission factor

$\%ER$ - overall emission reduction efficiency

$$\text{Operation} = 5 \frac{\text{hr}}{\text{day}} \times 7 \frac{\text{days}}{\text{week}} \times 52 \frac{\text{weeks}}{\text{year}} = 1820 \text{ hr/yr}$$

$$A = 1820 \frac{\text{hr}}{\text{yr}} \times 500 \frac{\text{kg}}{\text{hr}} = 910,000 \text{ kg/yr}$$

$$E = 910,000 \frac{\text{kg}}{\text{yr}} \times 5 \frac{\text{lb}}{\text{ton}} \times \frac{1 \text{ ton}}{2,210 \text{ lb}} \times \frac{1 \text{ ton}}{1,000 \text{ kg}} = 2.06 \frac{\text{ton}}{\text{yr}}$$

21. A poultry farm operates with 30,000 birds per cycle, completing 5 grow cycles per year. Using the provided emission factor for CH_4 (0.023 kg/head/year) and adjusting for grow time (38 days), what is the approximate annual methane emission from manure management?

- A. 138.75 kg/year
B. 276.30 kg/year
C. 359.49 kg/year
D. 521.60 kg/year

$$\begin{aligned} E_{\text{CH}_4} &= A \times EF = 30,000 \text{ heads} \\ &\times 38 \frac{\text{days}}{\text{grow}} \times 5 \frac{\text{grows}}{\text{year}} \\ &\times \frac{1 \text{ yr}}{365 \text{ days}} \times 0.023 \\ &= 359.49 \text{ kg/year} \end{aligned}$$

22. A poultry farm operates with 30,000 birds per grow cycle, with 5 cycles per year, applies manure to land without any treatment. Using and N_2O emission factor of 0.6 kg/head/year, adjusted for grow time (38 days), what is the estimated annual N_2O emission?

- A. 4.45 tons N_2O /year
B. 6.35 tons N_2O /year
C. 9.38 tons N_2O /year
D. 23.50 tons N_2O /year

$$\begin{aligned} E_{\text{N}_2\text{O}} &= A \times EF = 30,000 \text{ heads} \\ &\times 38 \frac{\text{days}}{\text{grow}} \times 5 \frac{\text{grows}}{\text{year}} \\ &\times \frac{1 \text{ year}}{365 \text{ days}} \\ &\times 0.6 \frac{\text{kg}}{\text{head} - \text{year}} \\ &= 9,378 \frac{\text{kg}}{\text{year}} \times \frac{1 \text{ ton}}{1000 \text{ kg}} \\ &= 9.38 \frac{\text{ton}}{\text{year}} \end{aligned}$$

23. Based on Item 8, what is the CO_2 -equivalent of the poultry farm's annual nitrous oxide emission, using IPCC GWP_{100} for N_2O = 298?

- A. 2,320 tons $\text{CO}_2\text{e}/\text{year}$
B. 2,795 tons $\text{CO}_2\text{e}/\text{year}$
C. 3,347 tons $\text{CO}_2\text{e}/\text{year}$
D. 4,168 tons $\text{CO}_2\text{e}/\text{year}$

$$\begin{aligned} \text{CO}_2\text{e} &= 9.38 \frac{\text{tons}}{\text{yr}} \times 298 \\ &= 2,795 \frac{\text{tons}}{\text{yr}} \end{aligned}$$

24. Based on Item 7, what is the CO_2e of the poultry farm's annual methane emission?

- A. 7.52 tons $\text{CO}_2\text{e}/\text{year}$
B. 8.99 tons $\text{CO}_2\text{e}/\text{year}$
C. 9.65 tons $\text{CO}_2\text{e}/\text{year}$
D. 10.13 tons $\text{CO}_2\text{e}/\text{year}$

$$\begin{aligned} \text{CO}_2\text{e} &= 359.49 \times 25 \\ &= 8,987 \text{ kg or } 8.99 \text{ tons} \end{aligned}$$

25. Oven method is a primary method of moisture determination. If after 3 days at 105°C the initial 20 g palay sample yields a final weight of 15 g, what is the moisture content in percent wet basis?

- A. 15
B. 20
C. 25
D. 33.3

$$\begin{aligned} \text{MC}_{\text{wb}} &= \frac{\text{mass of water}}{\text{mass of moist sample}} \\ &= \frac{20 - 15}{20} \times 100 \\ &= 25 \end{aligned}$$

26. If 2,200 lb of palay at 23% moisture content wet basis is loaded in the dryer and dried to 14%, calculate the amount of moisture removed.

- A. 100 kg
B. 105 kg
C. 220 lb
D. 230 lb

Moisture Removed

$$= W_i \left(1 - \frac{1 - MC_i}{1 - MC_f} \right)$$

$$= 2200 \text{ lb} \left(1 - \frac{1 - 0.23}{1 - 0.14} \right) = \text{---} \text{ lb}$$

27. Which of the following is the correct conversion equation from moisture content wet basis to moisture content dry basis?

- A. $MC_{db} = \frac{MC_{wb}}{100 - MC_{wb}}$
 B. $MC_{db} = \frac{MC_{wb}}{100 + MC_{wb}}$
 C. $MC_{db} = \frac{100 - MC_{wb}}{100 + MC_{wb}}$
 D. $MC_{db} = \frac{100 + MC_{wb}}{MC_{wb}}$

28. Paddy grain with 14% MC wet basis has approximately how much MC dry basis?

- A. 16.3
 B. 17.6
 C. 15
 D. 12.87

$$MC_{db} = \frac{MC_{wb}}{100 - MC_{wb}} \times 100$$

$$= \frac{14}{100 - 14} \times 100$$

$$= \text{---}$$

29. 200,000 lb of shelled corn at 24% MC wet basis are dried to 14% MC. Determine the weight of water removed.

- A. 23,256 lb
 B. 20,546 lb
 C. 2.56 kg
 D. 32.65 kg

Moisture Removed

$$= 200,000 \text{ lb} \left(1 - \frac{1 - 0.24}{1 - 0.14} \right)$$

$$= \text{---} \text{ lb}$$

30. A sample of corn has a moisture content of 11.8% on a wet basis. What is its moisture content on a dry basis?

- A. 10.5%
 B. 11.8%
 C. 13.4%
 D. 14.0%

$$MC_{db} = \frac{11.8}{100 - 11.8} \times 100 = \text{---} \%$$

31. Given:

Initial moisture content = 22%

Equilibrium moisture content = 14%

Actual moisture content = 18%

Calculate the moisture ratio.

- A. 0
 B. 0.5
 C. 0.75
 D. 1.0

$$MR = \frac{\text{actual MC} - \text{equilibrium MC}}{\text{initial MC} - \text{equilibrium MC}}$$

$$= \frac{18 - 14}{22 - 14} = \text{---}$$

32. If 20 kg of water is removed from a 110 kg paddy, what is the percentage moisture content of the paddy in wet basis?

- A. 12%
 B. 18%
 C. 22%
 D. 29%

$$MC_{wb} = \frac{\text{mass of water}}{\text{mass of moist sample}} \times 100$$

$$= \frac{20}{110} \times 100 = \text{---} \%$$

33. A product has 25% moisture content dry basis. What is the percentage moisture content of the product in wet basis?

- A. 10%
 B. 20%
 C. 30%
 D. 40%

$$MC_{wb} = \frac{MC_{db}}{1 + MC_{db}} \times 100$$

$$= \frac{0.25}{1 + 0.25} \times 100$$

$$= \text{---} \%$$

34. A solar dryer is used to reduce the moisture content of four tons of paddy rice from 25% to 15% wet basis. The amount of moisture removed is about:

- A. 0.47 ton
 B. 0.77 ton
 C. 0.97 ton
 D. 10.47 ton

Moisture Removed

$$= 4 \text{ tons} \left(1 - \frac{1 - 0.25}{1 - 0.15} \right)$$

$$= \text{---} \text{ tons}$$

Remember 1 ton = 1,000 kg
 1 kg = 2.20 lb

35. Amount of water needed to increase moisture content of 50 kg (fresh weight)

manure from 40% (wet basis) to 60% (wet basis).

- A. 25 kg
- B. 45 kg
- C. 75 kg
- D. 50 kg

$$\text{Moisture to Add} = 50 \left(\frac{1 - 0.40}{1 - 0.60} - 1 \right) = \text{--- kg}$$

36. The amount of moisture in the grain expressed as a percentage of the total weight of the sample.

- A. Moisture content dry basis
- B. Moisture content wet basis
- C. Moisture content
- D. Moisture removed

37. According to PAES 203:2000 - Moisture Content Determination for Rice and Corn, test samples shall be taken from different parts of the bag (for example, top, middle and bottom) by means of sack-type spear. How many samples are required for more than 100 bag lot?

- A. Each bag
- B. 10
- C. 30
- D. Square root of total number

38. Method of grain moisture determination based on actual extraction of water either by convection heating (oven method) or distillation.

- A. Primary method
- B. Secondary method
- C. Tertiary method
- D. Hybrid method

39. Measure of the ability of the machine to remove the hulls.

- A. Coefficient of hulling
- B. Hulling efficiency
- C. Coefficient of wholeness
- D. Wholeness efficiency

40. A component of a rice mill that removes the bran layer in the brown rice.

- A. Polisher
- B. Paddy grader
- C. Destoner
- D. Whitener

41. The expression used to indicate the amount of bran removed in the milling process.

- A. Bran recovery
- B. Degree of milling

- C. Head rice recovery
- D. Milling percentage

42. The percentage of grains that are free from foreign matter.

- A. Chaff
- B. Dockage
- C. Purity
- D. Immature grains

43. In conveying loose grains and bulk packages, the most appropriate materials handling equipment is

- A. Belt conveyor
- B. Bucket conveyor
- C. Screw conveyor
- D. Pneumatic conveyor

44. C

44. The angle measured from the horizontal at which product will start to move downwards over a smooth surface with gravity discharging the product is called

- A. Angle of discharge
- B. Angle of friction
- C. Angle of repose
- D. Hopper angle

45. Throughput capacity is the amount of grain which flows through a continuous-flow equipment in one hour. If the holding capacity is 6 ton, and it uses a 30-minute process, then throughput capacity is

- A. 3 ton
- B. 5 ton
- C. 6 ton
- D. 12 ton

$$\text{throughput} = \frac{6 \text{ ton}}{30 \text{ min}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times 1 \text{ hr} = \text{--- ton}$$

46. Given

Moisture Ratio = 0.2

Initial Moisture Content = 24%

Final Moisture Content = 16%

Estimate the equilibrium moisture content

- A. 12%
- B. 14%
- C. 28%
- D. 30%

$$0.2 = \frac{0.16 - x}{0.24 - x} \quad x = \text{--- \%}$$

Equilibrium moisture content - a moisture content where a material has no water activity

- d 47. Based on PAES 418:2002 – Primary Processing Plant for Fresh Fruit and Vegetable, the storage area of a processing plant shall have a capacity for temporary storage of raw materials for how many days?
- A. 5-10 processing days
B. 9-10 processing days
C. 5-7 processing days
D. 2-5 processing days

- b 48. What is the minimum ceiling height required for a fruit and vegetable primary processing plant according to PAES 418:2002?
- A. 2.0 m
B. 2.4 m
C. 2.8 m
D. 3.0 m

- c 49. An engineer is designing a primary processing plant with a target capacity of 4 tons of fresh vegetables per day. Based on PAES 418:2002, how much total floor area should be allocated at minimum for processing alone?
- A. 40 m²
B. 60 m²
C. 80 m²
D. 100 m²

Estimated Floor Area

$$= 20 \frac{\text{m}^2}{\text{ton}} \times 4 \text{ ton}$$

$$= \text{---} \text{m}^2$$

- d 51. On a psychrometric chart, heating and dehumidification process appears as a line that is:

- a. diagonal upwards to the right
b. diagonal downwards to the left
c. diagonal upwards to the left
d. diagonal downwards to the right

- d 52. Determine the maximum deflection in a uniformly distributed beam of length "L" carrying a uniform load "w" along its length. Modulus of elasticity in this beam is "E" and moment of inertia is "I".

- A. $5wL^4/438EI$ B. $5wL^3/834EI$
C. $5wL^3/348EI$ D. $5wL^4/384EI$

- d 53. This gutter is recommended for collecting cattle manure.
- a. gravity drains gutter b. scrape gutter
c. flush gutter d. step-dam gutter

- b 54. The introduction of substances not found in the natural composition of water that make the water less desirable or unfit for intended use
- a. water pollution b. contamination
c. impurification d. deterioration

- b 55. Precipitate resulting from coagulation or sedimentation of liquid waste.

- a. slurry b. sludge
c. effluent d. suspended solid

- b 56. In CPES rating, what is the equivalent score of very satisfactory?

- a. 82% to 89% b. 89% to 96%
c. 75% to 82% d. < 96%

- b 57. All projects shall be subjected to a minimum of ___ evaluations to be performed by the CPE during construction. Except for those projects with a duration of 90 calendar days and below which may be subjected to at least ___ visit.

- a. 1,2 b. 2,1 c. 2,3 d. 3,2

- d 58. In PAES 204:2015, the minimum purity of the mechanical rice thresher without sifter and with fan is: ___

- a. 50% b. 65% c. 80% d. 95%

- d 59. Teagasc is an organization based in ___, while the Codex Alimentarius was established by the FAO and ___.

- a. Dublin, Ireland and CGIAR
b. Dublin, Ireland and WFP
c. Dublin, Ireland and USDA
d. Dublin, Ireland and WHO

- d 60. When this type of curing is used, the pavement shall be cured initially with burlap or cotton mats, until after final set of the concrete or, in any case, for 12 hours after placing the concrete.

- a. white polyethylene sheet
b. waterproof paper
c. impervious membrane method
d. straw

61. In PAES 129:2002, this indicates the maximum load that can be successfully carried by the motor if it is to operate continuously and remain within a safe temperature range.

- a. service factor b. breakdown torque
✓ c. duty rating d. locked-rotor current

62. Arduino runs in _____ programming language and its IDE means:

- ✓ a. R, Internal Development Environment
b. Java, Interconnected Development Environment
✓ c. C/C++, Integrated Development Environment
d. Python, International Development Environment

63. In PAES 201:2015, the minimum heating system efficiency for biomass fuel heated air mechanical grain dryer for indirect-fired operation.

- a. 90% b. 75% c. 65% d. 50%

64. Its current lags the voltage at phase angle 90 degrees.

- ✓ a. Purely Inductive b. Purely Capacitive
c. Purely Resistive d. RLC Connection

65. In PAES 251:2018, the minimum main product recovery of corn mill is:

- a. 84% b. 86% c. 96% d. 64%

66. Any entity that produces agricultural and fishery machinery from prototyping, testing, commissioning, and selling. Usually, they produce agricultural and fishery machinery depending on the order and arrangement.

- a. producer b. assembler
✓ c. manufacturer d. fabricator

67. In PAES 201:2015, the minimum drying efficiency of mechanical grain dryer is:

- a. 90% b. 85% c. 80% d. 75%

68. In PAES 205:2000, it is the required straw length (cm) for mechanical rice thresher testing:

- ✓ a. 40-45 b. 45-50 c. 50-55 d. 55-60

69. In PAES 220:2004, the maximum mechanically damage kernel for peanut sheller is:

- a. 3.5 b. 3.0 c. 2.0 d. 7.0

70. In PAES 208:2000, the minimum shelling efficiency for corn sheller:

- a. 97.0 b. 99.8 c. 95.0 d. 99.5

71. Rice grain from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layer have been removed, but part of the lengthwise streaks of the bran layer may still be present on more than 40% of the sample grains

- a. Well-milled rice b. Regular milled rice
c. Over milled rice d. Undermilled rice

72. In PAES 202:2015, Heated air mechanical grain dryer – Methods of test, the grain to be used shall be homogeneous and the moisture content should be at least _____% for rice and corn.

- a. 18% b. 22% c. 24% d. 26%

73. In PAES 232:2017, precautionary measures in Wastewater Re-Use for Irrigation, which is false?

- a. Direct contact with wastewater shall be avoided
✓ b. Use of fine mist for sprinkler irrigation shall be used
c. Potable and wastewater lines shall not cross-connect
d. Irrigation with wastewater shall be stopped immediately when algal bloom occurs

74. It is the milled corn kernels with particle size between 1.1 mm to 1.19 mm.

- a. Grit #16 b. Grit #18 c. Grit #10 d. Grit #14

75. A structured and systematic technique for system examination and risk management

- a. HAZOP ☒ b. HACCP c. GMP d. SSOP

76. These are cracks or breaks across the annual rings of the wood during its growth caused by excessive bending of the tree due to strong wind.

- a. heart shakes ☒ b. cup shakes c. star shakes d. knots

77. In PAES 409:2002, how many milking operators are required for a milking herd having 11-20 animals without a milking machine?

- a. 1 ☒ b. 2 c. 3 d. 4

78. Doe/Ewe of 50 kg space requirement in slotted flooring.

- a. 0.7 sq. m. b. 0.9 sq. m. ☒ c. 1.1 sq. m. d. 2.5 sq. m.

79. The primary law governing blackbody radiation that quantifies the intensity of radiation emitted by unit surface area into a fixed direction from the blackbody as a function of wavelength for a fixed temperature.

- ☒ a. Stefan-Boltzmann Law
b. Newton's Law of Cooling
c. Fourier's Law
d. Planck's Law

80. Amount of heat added or removed measured in change of temperature without changing state.

- a. Latent Heat ☒ b. Sensible Heat
c. Specific Heat d. Innate Heat

☒ 81. In CPES rating, what is the percentage for Materials upon completion?

- ☒ a. 0.20 b. 0.15 c. 0.40 d. 0.30

82. Product of the coefficient of hulling and the coefficient of wholeness of grains, expressed in percent.

- a. Coefficient of wholeness b. Coefficient of hulling
c. Hulling Efficiency ☒ d. Fineness Modulus

83. The amount paid to project owners from Clean Development Mechanism (CDM) fund for carbon emission reduction projects.

- ☒ a. Carbon credit value b. Emission value
c. Carbon trading value d. Carbon emission value

84. What is the minimum methane content of biogas in order to operate biogas-fueled generators?

- a. 50% ☒ b. 65% c. 85% d. 90%

85. Type of biogas plant which have aerobic conditions in the upper layers and anaerobic processes occurring in the bottom layers, especially in the settled solids?

- a. Aerobic Biogas b. Anaerobic Biogas
☒ c. Facultative Biogas d. Activated Biogas

86. It is a wastewater, either treated or untreated, discharged from a reservoir, basin, or treatment plant.

- ☒ a. Effluent b. Influent
c. Turbid Water d. Hard Water

☒ 87. In PAES 421:2009, the tank shall have heating device able to heat a complete charge of bituminous liquid to ___ oC.

- a. 80 b. 120 ☒ c. 180 d. 200

☒ 88. In PAES 415:2001, under natural ventilating systems of greenhouse, large vent openings provide the most ventilation with total vent area that should be ___ - ___ of floor area.

- a. 15% - 30% b. 30% - 50%
c. 25% - 50% ☒ d. 15% - 25%

89. In PAES 421:2009, it is constructed to drain water from the subgrade and to collect surface water either from the roadway surface or adjacent.

- ☒ a. roadside ditch b. culvert
c. channel d. conduit

90. In PAES 408:2001, the minimum space requirement for carabao feedlot feeding shall be _____ m/animal.

- a. 0.5 b. 1.0 ☒ c. 0.75 d. 1.25

91. The maximum number of outlets for a 15-A circuit.

- ☒ a. 8-10 outlets b. 10-12 outlets
c. 14-15 outlets d. 6-8 outlets

92. Dissolved oxygen concentrations will be the lowest within a waterbody on _____.

- a. Afternoon b. Sunrise ☒ c. Sunset d. Noon

93. In PAES 401:2000, heaters shall be installed in creep areas to provide newborn pigs its required temperature of _____ OC - _____ OC until they are 3 days old.

- a. 32-35 ☒ b. 29-35 c. 30-35 d. 27-35

94. It is an environmental-friendly refrigerant.

- ☒ a. CFC ☒ b. HFC c. HCFC d. FC

95. It is the programming language of Arduino microcontrollers

- a. R b. Python ☒ c. C/C++ d. Java

96. In Arduino programming, the meaning of IDE is:

- ☒ a. Integrated Development Environment
b. Internal Development Environment
c. International Development Environment
d. Interconnected Development Environment

97. The Feed-in-Tariff (FIT) of wind energy in the Big SHOW is:

- a. 6.63 PHP/kW-hr
b. 9.68 PHP/kW-hr
c. 5.90 PHP/kW-hr
☒ d. 8.53 PHP/kW-hr

98. It is the gas produced from agricultural waste, which is odorless, tasteless, colorless and flammable.

- a. CH₄ ☒ b. CO c. O₂ d. CO₂

99. All are products of pyrolysis except:

- ☒ a. Vinegar b. Tar
c. Oil d. Syngas

100. It is a type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing.

- a. friction type
☒ b. abrasive type
c. rubber roll type
☒ d. cone "cono" type

**"You are braver than you believe,
stronger than you seem,
and smarter than you think."**